

Storage Cost Internalization in Bitcoin’s Fee Market

The Bitcoin Block Space Problem (program subtitle)

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Abstract

Bitcoin has a market price for block space, but no explicit market price for long-lived resource consumption. This paper measures one of those resources — replicated storage — and asks how much of its modeled cost is covered by transaction fees. We define a **Storage Cost Coverage Ratio (SCCR)** — the ratio of transaction fees paid (USD) to the estimated lifetime storage cost borne by full nodes (USD) — and measure it against live fee-history data.

The headline is a range, not a point. Our best available measurement of the replication factor N (full-node count) is a **primary-source lower-bound census ($\geq 32,000$ known addresses via Bitcoin Core `getnodeaddresses`)** — the address-manager cap, not a complete enumeration — while independent estimates span $\sim 10\text{K}$ – 100K reachable nodes. Because SCCR is exactly inverse-linear in N , the result is stated in four lines (exact figures appear in §5):

- **Dated baseline band (Aug 02–15): ≈ 0.22 coverage at $N = 32\text{K}$,** with $\sim 100\%$ of sampled blocks below $1\times$ storage cost.
- **Live series (Sep 07): 0.40 at $N = 32\text{K}$, 94.37% of blocks below $1\times$** — the coverage ratio is **fee-market-driven and time-varying**, not a fixed point.
- **Dynamic finding: SCCR moved $0.16 \rightarrow 0.45 \rightarrow 0.40$ across a 22-point daily series (2026-08-02 $\rightarrow$ 2026-09-07)** measured live in `data/sccr_history.json` — dynamic measurement strengthens the paper.
- **Model uncertainty: depends strongly on the replication factor N** — the average spans ~ 0.71 at $N = 10\text{K}$ to ~ 0.07 at $N = 100\text{K}$; this N -band range (~ 0.07 – 0.71) is a *different* uncertainty from the dated/live observations above and carries the dominant risk.

A joint Monte Carlo over N , C , T , and price (current band, 10,000 draws) gives a P5–P95 interval of 0.07 – 0.47 , median 0.17 , with 99.9% of draws below the $1\times$ threshold; the share of sampled blocks below $1\times$ ranges ~ 79 – 100% depending on the true N . We reconcile two cost models that previously disagreed by $16.4\times$ (dimensionless), document the correction transparently (model-spec.json v2.0.0), and state results to the precision the evidence licenses: **external reproduction is pending (D5)**. The framework is reproducible, falsifiable research.

Hypothesis: Bitcoin’s fee market efficiently allocates scarce block space, but may not fully internalize every long-lived resource cost created by confirmed transactions.

1 Introduction: Congestion Pricing vs. Permanence Cost

The one-sentence frame (full treatment in `problem_statement.md`):

Bitcoin’s fee market prices competition for inclusion in the next block, but it does not explicitly price the long-term resource costs of permanently recorded blockchain data.

The fee market solves a short-term optimization problem extremely well: during congestion, higher fees win block space. This is **congestion pricing**, with a horizon of roughly one block (~ 10 min). Storage cost is structurally different:

- **One-time payment** (the fee, USD/block) vs. **recurring cost** (long-term replicated storage across the network, USD/(node·yr))
- **Payer chooses** to pay vs. **node operators bear** the cost involuntarily
- ~ 10 -min horizon (next block) vs. **indefinite horizon** (data persists in blockchain history)

Scope: this paper measures the **storage leg** of a broader resource-pricing question. Bandwidth, validation, and UTXO-maintenance legs now carry v1 bounds/measurements (§5 5.5–5.7) with the residual program in §7.

Why storage? It is not the “most important” resource — it is simply the first measurable one. Storage qualified first because its cost leg (C , N , T) and its fee leg (block fees, USD) are both estimable from primary sources. Every other resource is a named research hypothesis, not a measured result.

Storage $\neq$ state. SCCR measures the cost of replicated *history* — the permanent record of confirmed blocks retained by full nodes. It does *not* measure the **UTXO set**, the live ledger state every node maintains in RAM and index structures. State permanence is a separate resource with its own accounting leg (UCIR, §7 / roadmap Phase II).

1 $\times$ is a descriptive calibration point, not a normative target. Stated at the outset so no reader mistakes the paper for a claim that fees *should* cover 100% of modeled storage cost. The paper measures whether they do; it prescribes nothing.

2 What We’re NOT Saying

- $\times$ Not proposing a fix
- $\times$ Not claiming the externality is economically significant at current volumes (it may not be)
- $\times$ Not claiming the SegWit discount was a mistake (it solved transaction malleability)
- $\times$ **Not arguing that Bitcoin is ‘broken’** — the fee market solves block-space allocation well; whether it internalizes long-lived resource costs is an empirical question
- $\times$ **Not claiming fees *should* cover 100% of storage cost** — the 1 $\times$ threshold is a descriptive calibration point, not a normative target (§1, §5.4)
- $\checkmark$ Just framing the question clearly and measurably

3 The SegWit Pricing Structure

SegWit (BIP 141, activated August 2017) introduced block weight: witness data counts **1 weight unit (WU) per byte**, non-witness data **4 WU per byte**. This **4:1 (dimensionless) discount** reduces the marginal block-space cost of witness-resident data relative to non-witness data. Years later, inscription protocols (Ordinals, BRC-20, Runes) took advantage of that pricing structure. Two precise caveats (full treatment in `bip141_analysis.md`):

1. The discount applies to **all** witness data — SegWit financial transactions benefit identically. The “accident” is that it also subsidizes data-bearing constructions, not that SegWit was designed for them.
2. Whether the discount appropriately reflects the long-term resource costs of the data it makes cheaper is **part of the research question**, not a foregone conclusion.

4 The Model

4.1 Canonical specification

All quantities, units, and formulas live in `research/model-spec.json` (v2.0.1) — a single canonical source that every script imports. No script redefines a model constant.

Quantity	Symbol	Units	Value
Annual node cost	C	USD/yr	925
Replication factor	N	nodes	32,000
Storage horizon	T	yr	10
Average block size	B_{block}	bytes	1,500,000
Blocks per year	R_{blocks}	blocks/yr	52,596
Total block bytes/yr	$B_{\text{all,yr}}$	bytes/yr	7.8894×10^{10}
Cost per byte per year	c_b	USD/(byte·yr)	1.17246×10^{-8}
Lifetime storage cost/node/block	L	USD/block	0.175869
Network lifetime cost/block	L_{net}	USD/block	5,628
Storage Cost Coverage Ratio	SCCR	dimensionless	0.2228 (167 blocks, live 2026-08-02); ~ 0.29 (156 blocks)

$$\text{SCCR} = \frac{\text{Fees Paid (USD/block)}}{\text{Modeled Lifetime Storage Cost (USD/block)}} = \frac{\text{fee}_{\text{USD}}}{L_{\text{net}}} = \frac{\text{fee}_{\text{USD}}}{C \cdot T \cdot N / R_{\text{blocks}}}$$

Definition of “fees paid” (used once, everywhere). Throughout this paper, *fees paid* means **total transaction fees of the block in USD**: $\text{fee}_{\text{USD}}(\text{block}) = \text{avgFees}(\text{block}) \times \text{USD/BTC price}$, where *avgFees* is the block’s total transaction fees in satoshis from the live `fee_history` capture (mempool.space 24h block-fee history, into `captured-data/bsahi.db`). It **excludes the block subsidy** (miner revenue, not a fee) and is **not** a fee *rate* (sat/vB). All three independent implementations (JS, Python, standalone C) compute $\text{fee}_{\text{USD}} = \text{avgFees}/10^8 \times \text{USD}$ identically. The denominator it is compared against is $L_{\text{net}} = C \times T \times N / R_{\text{blocks}}$ (USD/block).

Design principle: c_b is **horizon-free** (C / annual bytes). The horizon T enters **only** through $L = c_b \times B \times T$. This corrects the v1.0.0 implementation, which applied T twice (see §6).

Explicit notation (hidden assumptions made visible). The canonical quantities bundle four assumptions the model treats as scalars; each is stated explicitly so readers see exactly what is and is not modeled. None changes the arithmetic of §5:

1. C is a **bundled cost, not pure storage**. $C = C_{\text{storage}} + C_{\text{bandwidth}} + C_{\text{misc}}$ — the \$925/yr annual node cost bundles disk/SSD, bandwidth, and electricity/hardware depreciation (component sum 166.67 + 600 + 157.68). The model treats C as a single bundled scalar, so the headline ratio is strictly a **storage-and-hosting coverage ratio**. The bandwidth-vs-storage

decomposition is a documented future refinement — exactly the RCIR and BCIR legs (§7 / roadmap Phase II–III).

2. c_b is **time-dependent, not stationary**. The canonical c_b is a point-in-time ratio $c_b(t) = C(t)/B_{\text{year}}(t)$. $B_{\text{year}}(t)$ depends on block fullness: 95%-full blocks produce $\sim 7.5 \times 10^{10}$ bytes/yr vs $\sim 5.5 \times 10^{10}$ at 70%-full blocks, moving c_b (and SCCR) even with C , N , T fixed. SCCR as measured is therefore a **point-in-time estimate over its capture window** (§5.1).
3. N is a **heterogeneous vector, not a scalar**. The model writes $L_{\text{net}} = L \cdot N$; the explicit forward notation is $L_{\text{net}} = \sum_i L_i$, where each node class i (archival, pruned, exchange, research/indexer) carries different storage behavior and hence a different per-node lifetime cost L_i . This is precisely the archival-vs-pruned measurement gap documented in the companion note `archival-vs-pruned-note.md`: the census measures reachable count N ($\geq 32\text{K}$, a lower bound), not the retention distribution.
4. **Storage $\neq$ state (UTXO)**. SCCR prices replicated *history* — the permanent record of confirmed blocks — not the UTXO set, the live ledger *state* nodes maintain in RAM and index structures. State permanence is a separate resource with its own accounting leg (UCIR, §7 / roadmap Phase II).

4.2 The marginal-inscription attribution (secondary)

The inscription-externality branch uses a **marginal attribution**: $c_{b,\text{insc}} = C/(\text{inscription bytes/yr}) = 1.92573 \times 10^{-6}$ USD/(byte·yr). This differs from c_b by a factor of **164.4 $\times$ (dimensionless)** — not an error, but two different denominators (all block bytes vs. inscription-only bytes). The paper headline uses the block-average c_b ; the marginal figure appears only in the inscription-externality analysis. The 16.4 $\times$ (dimensionless) figure reported in earlier versions was this 164 $\times$ (dimensionless) denominator gap **divided by** the 10 $\times$ (dimensionless) bug — see §6.

Why average, not marginal? c_b is an *average* (accounting) attribution: total node cost divided by total bytes. The 164 $\times$ (dimensionless) marginal branch ($c_{b,\text{insc}}$) attributes the same cost to inscription bytes only — a *marginal* (optimization) view. Both are valid; they answer different questions. Average cost is the right tool for **accounting** — “what share of the network’s modeled hosting bill does the fee market cover?” — which is this paper’s question. Marginal cost is the right tool for **optimization** — “what does one more byte cost the network?” — which is the inscription-externality branch’s question. The paper uses the average for its headline (accounting) and documents the marginal variant explicitly, rather than burying the choice.

5 Findings

5.1 The Storage Cost Coverage Ratio

Measured from live `fee_history` captures at node count $N = 32,000$ — the best-available **primary-source lower-bound census** ($\geq 32,000$ known addresses via Bitcoin Core `getnodeaddresses`; §5.4). The headline, in three lines (exact numbers in the tables below):

- **Representative live measurement:** ≈ 0.22
- **Observed band:** $\approx 0.22\text{--}0.29$ (across captures at $N = 32\text{K}$)
- **Model uncertainty:** depends strongly on the replication factor N — the true- N band ($\sim 0.07\text{--}0.71$, §5.4) is a *different* uncertainty from the observed-sample band above

Two snapshots, dated explicitly:

Metric	Dated capture (2026-08-01)	Live capture (2026-08-02)
Blocks sampled	156	167
Average SCCR (dimensionless)	~ 0.293	0.2228
Min / Max (dimensionless)	~ 0.058 / ~ 1.537	~ 0.058 / ~ 0.832
Blocks below $1\times$	98.7% (154/156)	100.0%

The ratio moves with the fee market; the single source of truth is `research/model-spec.json` (v2.0.1, canonical); all surfaces must read the live value from `node tools/research/storage-ratio.js`, never a hardcoded figure. The v2.0.0 N=60K-era values (0.1719 / 0.1535) are superseded.

Fee-regime + node-count dependence: the ratio tracks both the fee market and the replication factor. Under the earlier N=60K assumption the average was 0.156–0.172 (dimensionless) with 100% below $1\times$; at the primary-source lower-bound census N=32K it rises to ~ 0.22 – 0.29 with ~ 98.7 – 100% below $1\times$ (a few high-fee blocks exceed coverage in the dated capture). The headline is a *distribution over time and parameters*, not a point.

5.2 The inscription externality (marginal branch)

At 100,000 inscriptions/month (~ 400 bytes each, in witness):

Metric	Value
Cost per byte per year (marginal)	1.92573×10^{-6} USD/(byte·yr)
Lifetime storage cost per inscription (10 yr)	\$0.00770 USD/inscription
New 10-yr storage liabilities created per year at 100K/mo	$\sim \$9,240$ USD/yr spread across all nodes
Steady-state annual externality (amortized)	$\sim \$924$ USD/yr spread across all nodes

Unit note (corrected): \$9,240 is the *undiscounted 10-year liability of one year’s inscriptions* (USD/yr), not an annual cost. Amortized over the 10-yr horizon it is $\sim \$924$ /yr. This was previously labeled “annual unpriced externality” — a T/1 conflation of the same shape as the fixed $10\times$ bug, in label only.

Pruning-consistency note: this externality is structurally real but **not economically significant at current volumes** — per-node it is well under \$1/yr. The contribution of this paper is the reproducible ratio framework, not the magnitude.

5.3 Sensitivity

Recomputed at the live capture (2026-08-02, 167 blocks; baseline SCCR = 0.2228 (dimensionless) at N=32K, C=\$925/yr, T=10 yr). The table’s parameter ranges bracket the canonical values:

Parameter	Value	Avg SCCR (dimensionless)
Node cost, C (USD/yr)	600 / 925 / 1400	0.343 / 0.223 / 0.147
Node count, N (nodes)	16,000 / 32,000 / 100,000	0.446 / 0.223 / 0.071
Storage horizon, T (yr)	5 / 10 / 15	0.446 / 0.223 / 0.149
Avg block size, B (MB)	1.0 / 1.5 / 2.0	0.223 (invariant)

BTC price (the dominant omitted driver — verified in independent implementation): the ratio is linear in price. At the live $N=32K$ baseline (capture price $\approx \$63,018$ USD/BTC): $\$30K \rightarrow 0.106$, $\$45K \rightarrow 0.159$, $\$62.9K \rightarrow 0.222$, $\$100K \rightarrow 0.354$, $\$200K \rightarrow 0.707$, $\$500K \rightarrow 1.768$.

Caveat — price and fee level are not independent shocks. The table above varies the BTC price holding the fee level (sat/vB) fixed, but in historical data fee levels and price co-move: bull markets raise both the price *and* on-chain fee demand, so a price rise is typically accompanied by a fee-level rise, compounding the ratio’s increase; a price fall in a demand crash carries fee levels down with it. The single-lever rows are *ceteris-paribus* isolations for the model’s arithmetic, not forecasts along a realized price path; the joint direction of price-and-fee co-movement is exactly the kind of dynamic question the v3.0 agenda (companion `future-directions-v3.md`) leaves open.

Structural property: SCCR is invariant to block size — $c_b \propto 1/B$ while $L \propto B$, so B cancels. Caveat: this invariance follows from attributing *all* node cost per byte then multiplying back by bytes — it reflects that the model is size-agnostic (contains no size-dependent economics), not that size is economically irrelevant.

5.4 The knife-edge: node count and the strong claim

The SCCR is homogeneous of degree 1 in its scale parameters: $SCCR \propto (\text{fee} \times \text{price}) / (C \times T \times N)$. Two thresholds matter (derived and verified in the independent C implementation):

- The average SCCR crosses 1.0 at $N \approx 10,300$ nodes (or BTC $\approx \$366,000$) — dated v2.0.0-era reference values (baseline 0.1719 at $N=60K$)
- The “100% of sampled blocks below $1\times$ ” claim breaks at $N \approx 49,200$ nodes (or BTC $\approx \$76,700$) — the highest-fee sampled block (13.75M sats, height 960469) sits at $0.82\times$ coverage under the old $N=60K$

Live recompute (2026-08-02 baseline, SCCR = 0.2228 at $N=32K$): the average now inverts at $N \approx 7,130$ nodes or BTC $\approx \$283,000$; the “100% below $1\times$ ” break on the dated capture is unchanged at $N \approx 49,200$. Both thresholds are reported; model-spec v2.0.1 retains the v2.0.0-era values (10.3K / $\$366K$).

The node census — a primary-source lower-bound census ($\geq 32,000$ known addresses via Bitcoin Core `getnodeaddresses`), not a full enumeration. We replaced the 60K assumption with data from a live Bitcoin Core node: a `getnodeaddresses` RPC query returned **32,000 known addresses** — the RPC maximum, meaning the node’s address manager is saturated at the cap and the true reachable set is *at least* 32K. At census time the node also reported 8 live outbound P2P connections (2026-08-02), small relative to the 32K known-address lower bound. **We do not call this a complete census:** 32K is an artifact of the RPC cap, not a measured enumeration, and independent estimates span $\sim 10K$ – $100K$. The paper therefore reports the ratio as a **range across the true-N band**, with 32K as the best-available lower-bound estimate.

Consequence of the real $N=32,000$ (recomputed, dated 156-block capture):

Metric	At $N=60K$ (old assumption)	At $N=32K$ (primary-source lower-bound census)
Average SCCR (dimensionless)	0.156–0.172	~ 0.293
Max per-block ratio (dimensionless)	0.820	1.537
Blocks below $1\times$	100.0%	98.7% (154/156)

This is a substantive finding, not a cosmetic one. With a defensible node count, fees cover *more* of the modeled storage cost than the 60K assumption implied ($\sim 29\%$ vs $\sim 17\%$), and a handful of high-fee blocks now *exceed* $1\times$ coverage. The direction of the headline is unchanged

— most blocks are still below $1\times$ — but the strong form (“100% below $1\times$ ”) does **not** survive the primary-source lower-bound census on the dated capture. **The defensible claim:** $\sim 22\text{--}29\%$ average coverage at $N=32K$, $\sim 98.7\text{--}100\%$ of sampled blocks below $1\times$, and $\sim 0.07\text{--}0.71$ average across the true- N band ($10K\text{--}100K$). ($1\times$ remains a descriptive calibration point, not a normative target — §1.)

Joint Monte Carlo on the headline (`research/sccr_monte_carlo.py`, 10,000 samples, $N \sim \text{Tri}(10K, 150K, \text{mode } 60K)$, $C \sim \text{Tri}(\$500, \$2000, \text{mode } \$925)$, $T \sim \text{Tri}(5, 30, \text{mode } 10)$, $P \sim \text{Tri}(\$30K, \$120K, \text{mode } \$62.9K)$):

Quantile	SCCR (dimensionless)
P5	0.034
P25	0.062
P50	0.099
P75	0.163
P95	0.338
Share below $1\times$	99.8%

The median is below the deterministic point estimate because the node-count distribution’s right tail (median $N \approx 86K$) dominates. The result is robust in distribution.

Current- N -band Monte Carlo (`research/sccr_monte_carlo_range.py`, 10,000 draws, $N \sim \text{Tri}(10K, 100K, \text{mode } 32K)$ — the paper’s stated uncertainty band with the census as mode; $C \sim \text{Tri}(\$600, \$1400, \text{mode } \$925)$, $T \sim \text{Tri}(5, 15, \text{mode } 10)$, $P \sim \text{Tri}(\$30K, \$120K, \text{mode } \$63,018)$; anchored at the live baseline and cross-checked against the frozen capture):

Quantile	SCCR (dimensionless, live anchor)
P5	0.071
P25	0.117
P50	0.173
P75	0.260
P95	0.467
Share below $1\times$	99.9%

The current-band median (0.17) sits near the $N=32K$ point estimates (0.22–0.29), and the P5–P95 interval (0.07–0.47) brackets the deterministic N -band range ($\sim 0.07\text{--}0.71$): the confidence interval and the range are two views of the same N -driven uncertainty. The old- N -band result above (99.8% below $1\times$) remains as the historical conservative check.

5.5 The Bandwidth Leg (v1 analytical bound)

(Added 2026-08-04.) The storage leg prices *stored* bytes; the bandwidth leg prices the **marginal propagation** of those bytes: every full node — pruned or archival — must download every byte of every block. Quantities live in `model-spec.json` v2.1.0 (`cost_per_gb` input; derived `bw_GB_yr`, `bw_cost_per_year_node`, `bw_cost_per_year_net`, `bw_insc_incr_node`); the formula is $B \times \text{replication} \times \$/\text{GB}$:

Quantity	Value	Meaning
$\text{bw_GB_yr} = B_{\text{all,yr}}/1\text{e9}$	78.894 GB/yr	full-chain bytes each node downloads
cost_per_gb (input)	\$0.05/GB	retail bandwidth proxy (AWS egress $\sim$ \$0.09/GB, Hetzner $\sim$ \$0.05/GB)
$\text{bw_cost_per_year_node}$	\$3.94/yr per node	marginal full-chain propagation, one node
$\text{bw_cost_per_year_net}$	$\sim$\$126K/yr	$\times N = 32\text{K}$ lower-bound census
bw_insc_incr_node	\$0.024/yr per node	inscription-incremental share ($480 \text{ MB/yr} \times \$0.05/\text{GB}$) —

Honest framing: this is an **analytical bound, not a measurement**. Real node bandwidth bills are overwhelmingly flat-rate (residential/colocation unmetered), so $\$/\text{GB}$ is a *marginal economic proxy*, not a typical bill. The full-chain marginal cost is $\sim$ \$3.94/yr per node ($\sim$ \$126K/yr network at $\geq 32\text{K}$ nodes), and the inscription-incremental slice is $\sim$ \$0.024/yr per node — small in both frames, in the same direction as the pruning analysis verdict ($\$0.02$ – $\$0.05/\text{yr}$ unavoidable per node).

5.6 The Validation Leg (v1 order-of-magnitude survey)

(Added 2026-08-04; full survey: `research/validation-cost.md`) Validation is the CPU work every node pays for every block since genesis — PoW-header check (trivial), block rules, and **signature verification** (the dominant term) — and it is the only leg strictly unavoidable at any replication scale (storage can be pruned; validation cannot be skipped). The v1 bound, **order-of-magnitude only** (band $\approx 0.5\times$ – $5\times$):

Term	Value
Blocks/yr (R_{blocks})	52,596
Sig checks per block / throughput	~ 3 – 10K sigs/block vs ~ 10 – 30K sigs/s (modern hw, libsecp256k1)
Validation CPU per block	~ 0.1 – 1 s (cross-checked by initial-sync delta: 6 – 24 h / $\sim 1\text{M}$ blocks)
Validation CPU per node per year	~ 1.5 – 15 h
Validation cost per node per year	$\sim \$0.5$ – $\$5$ (central $\approx \$1$ – $2/\text{yr}$ at $\$0.10$ – $0.50/\text{CPU-h}$)
Network-wide ($N = 32\text{K}$)	$\sim \$16\text{K}$ – $160\text{K}/\text{yr}$

Falsifiable claim (v1): *validation cost per full node per year is $< \$100$ — bounded from above by the entire node budget $C = \$925/\text{yr}$, central estimate $\sim \$1$ – $2/\text{yr}$.* Falsified by a measured benchmark showing $\geq \sim 200 \text{ h/yr}$ steady-state validation CPU, or a hardware census showing validation is a *binding* provisioning constraint. Literature anchored on Tschorsch & Scheuermann (2016, IEEE COMST), Delgado-Segura et al. (2018, UTXO analysis), Bitcoin Core’s `src/bench` suite. **Verdict:** validation is cheap per block, so cheap per node; its network total ($\sim \$16\text{K}$ – $160\text{K}/\text{yr}$) is ~ 3 orders of magnitude below the storage leg’s modeled network burden — which is itself the finding, not an assertion.

5.7 The UTXO Leg (v1 measurement)

(Added 2026-08-04.) `getblockstats` $\rightarrow$ `utxo_size_inc` — the net byte delta of the UTXO set after each block — was already captured by agent-06 but **dropped at DB write** (no `block_stats` column). Fixed end-to-end: column added to `tools/db/schema.sql`, wired through `tools/db/init.js` (`insertBlockStats` + idempotent `ALTER TABLE` migration), live path in `tools/data-engineering/spool-consum`

and a backfill (`tools/db/backfill-block-stats.js`) populated history from spool + raw captures. Measured v1 result (2026-08-04, `captured-data/bsahi.db`):

Metric	Value
<code>block_stats</code> rows with <code>utxo_size_inc</code>	165 unique heights (backfill: 351 writes, 119 capture files)
Avg net UTXO-set delta per block	~29.9 KB/block (range incl. genesis’s 117 B artifact)
Max height covered	671,460
Live path	verified (handler test: <code>height→utxo_size_inc</code> persisted)

Status: the leg is now *measured* (the resource’s size per block is in the queryable DB, growing live). What remains is the *pricing* step — attributing a \$/byte/yr cost to the UTXO delta — which stays in §7 future work until then.

6 Internal Validation, Correction, and Reconciliation (v2.0.0)

Internal validation identified an inconsistency in the SCCR implementation, traced it to a **duplicate time-horizon term**, corrected the implementation, regenerated all reported values, and confirmed the qualitative conclusions unchanged. Four labeled steps: **Bug** → **Fix** → **Results** → **Reconciliation**.

6.1 The bug

`methodology.json` v1.0.0 and `storage-ratio.js` applied the horizon T twice — once dividing the denominator of `costPerBytePerYear` ($C/(B_{\text{all,yr}}/T)$) and again in the lifetime-cost product ($\text{bytes} \times c_b \times T$). This inflated modeled storage cost (USD/block) — and deflated the ratio — by exactly $10\times$. The error affected the magnitude, not the logical structure of the model.

6.2 The fix

c_b is defined horizon-free ($C/\text{bytes-per-year}$); T enters only through $L = c_b \times B \times T$. This is the canonical v2.0.0 definition, pinned in `research/model-spec.json`; no script may reintroduce T into c_b .

6.3 Results

Quantity	Before (v1.0.0 formula, N=60K)	After (v2.0.0, N=60K)
Cost per byte per year (USD/(byte·yr))	1.17247×10^{-7}	1.17247×10^{-8}
Lifetime storage cost/node/block (USD/block)	\$1.759	\$0.176
Network lifetime cost/block (USD/block)	\$105,522	\$10,552
Average SCCR (dimensionless)	0.0172	0.1719
Blocks below $1\times$	100%	100% (unchanged)

Why the conclusion survived the correction. Although the correction increased the estimated SCCR by $\sim 10\times$ on a same-capture basis (and $\sim 11.5\times$ across the as-reported headlines v1.0.0

0.0149 $\rightarrow$ v2.0.0 **0.1719**), every sampled block still remained below the modeled full-cost threshold ($1\times$) under the then-assumed node count ($N=60K$). At the primary-source lower-bound census $N=32K$ the average rose further to ~ 0.225 on the 2026-08-02 live re-measure (168 blocks), with $\sim 99\%$ of sampled blocks still below $1\times$. The magnitude moved by an order of magnitude; the *direction* of the finding did not.

6.4 Reconciliation

The two cost models (`storage-ratio.js` and `utxo_cost_model.py`) disagreed by $16.4\times$ (dimensionless). This decomposed as $164\times$ (dimensionless) denominator gap $\div 10\times$ (dimensionless) bug = $16.4\times$. The models measure different quantities (block-average vs. marginal-inscription attribution); the gap is documented, not erased. See `verification_appendix.md`.

6.5 Reproducibility

Every quantity in this paper is regenerated from `research/model-spec.json` (v2.0.1) by three independent implementations (JS, Python, standalone C); no script redefines a model constant, and the full capture log is retained in `captured-data/bsahi.db`. A frozen input capture and a cross-check script ship in `research/reproduce/` (see README).

7 Limitations and Future Work

Limitations:

1. The node count ($\geq 32K$ from the primary-source lower-bound census) is a **lower bound**, not a complete enumeration. SCCR is inversely proportional to N : at the live baseline the average inverts above $1\times$ only below $\sim 7.1K$ nodes, and the “100% below $1\times$ ” claim breaks below $\sim 49K$ nodes at the dated capture (§5.4).
2. **Node costs are homogeneous — and bundled.** Hardware/bandwidth/electricity vary by geography and operator; the model treats them as one scalar $C = \$925/\text{yr}$. Because C is bundled ($C = C_{\text{storage}} + C_{\text{bandwidth}} + C_{\text{misc}}$, §4.1), the headline ratio is strictly a **storage-and-hosting coverage ratio**, and the bandwidth-vs-storage decomposition is a documented future refinement (RCIR and BCIR legs).
3. 10-year horizon is an assumption; pruning shortens actual retention, permanent storage extends it.
4. **No discounting.** A one-time fee (USD/block) is compared against an undiscounted 10-yr storage-cost sum; discounting the liability at $r = 5\%/\text{yr}$ ($8\%/\text{yr}$) reduces the present value by $\sim 27\%$ (45%).
5. Bandwidth is included in the fixed node cost (C) yet marginal bandwidth-propagation cost is excluded from the storage leg (fixed-vs-marginal distinction).
6. Marginal vs. average attribution changes the per-byte cost by $164\times$ (dimensionless) — the choice is explicit and documented.
7. **Why storage at all?** The paper does not claim storage is Bitcoin’s most important resource — **it is simply the first measurable one**. Bandwidth, validation, UTXO, relay, and indexer serving are named research hypotheses with their own metrics (roadmap §4), none yet measured.

Future work (the resource-pricing program; statuses updated 2026-08-04): UTXO leg — v1 DONE (measurement: `utxo_size_inc` persisted end-to-end, §5.7; ~ 29.9 KB/block avg, 165 backfilled heights); remaining: $\$/\text{byte}$ pricing of the UTXO delta. Bandwidth leg — v1 DONE

(analytical bound: model-spec v2.1.0, \$3.94/yr per node marginal propagation, §5.5); remaining: measured per-node bills. Validation leg — v1 STARTED (OOM survey, `validation-cost.md`, <\$100/yr/node central \$1–2/yr, §5.6); remaining: pinned Core benchmark + hardware census. Node distribution (`node_geo`, 224 rows); BIP-110 pre/post measurement protocol if ever signaled; v3.0 economic-dynamics program (companion `future-directions-v3.md`, agenda summary §10).

7.1 What would falsify this framework?

(Added 2026-08-03, post-advisor review — see `docs/decisions/2026-08-02-publication-decisions.md`.)

A framework that cannot specify its own failure conditions is not a framework; it is a posture. We state, in advance, the observations that would force us to retract or substantially revise the claims in this paper. They operate at two levels — the **measurement** (SCCR, this paper) and the **framework** (the RIR family and the “no single resource market” thesis; outline in `research/framework-paper-outline.md`).

Two external outcomes — never conflated (added 2026-08-03, third-reviewer refinement). An external party’s run of the reproduction protocol can end in exactly two ways, and they are categorically different:

- **“Failed to reproduce”** — an independent party following the published protocol from a clean clone gets a materially different number: not the published avg 0.2186, not the 0.07–0.71 band, or a per-block mismatch that cannot be reconciled. This is a **genuine falsification of the measurement**: the number is not trustworthy as stated, and the paper’s headline falls until the discrepancy is reconciled. This is the only external-outcome class that blocks submission (falsifier 1 below; D5).
- **“Reproduced the number, disagrees with framing/assumptions”** — an independent party runs the protocol, confirms the arithmetic (avg 0.2186, the band, the below-1× share), but disputes a modeling choice: the $C = \$925/\text{yr}$ bundling, the $T = 10$ horizon, storage-as-the-first-resource priority, or whether an unpriced-but-avoidable cost is an externality at all. This is **NOT a falsification**. It is honest scientific disagreement about documented assumptions — the paper states these choices as assumptions (§7, items 1–7) and anticipates exactly this class of challenge. It is folded into future revisions the way the Liu et al. (2021) prior work was reconciled (§8.2): acknowledged, engaged, absorbed into the next version. The measurement stands.

The two outcomes must never be conflated. A framing disagreement is **feedback, not a failed reproduction**: it is recorded in the community-feedback triage (`research/community-review-plan.md` §4 → `research/community-feedback.md`) and addressed in the next revision, while the reproduced number stands. A failed reproduction is a claim about the number itself and is handled under falsifier 1.

Falsifiers of the SCCR measurement:

1. **Independent implementations cannot reproduce the ratio.** The framework ships a frozen capture plus three independent implementations (JS/Python/C, verified agreeing per-block). If an uninvolved party running the protocol on the same inputs cannot reproduce $\text{SCCR} \approx 0.22$ (or the discrepancy cannot be reconciled), the measurement is not trustworthy and the paper’s headline falls. This is the external-reproduction critical path (`research/reproduce/external-reproduction.md`).
2. **A better storage-cost model reverses the conclusion.** The denominator ($C \cdot T \cdot N/R_{\text{blocks}}$) rests on a homogeneous node-cost model, an assumed $T = 10$ horizon, and the $\geq 32\text{K}$ primary-source lower-bound census. If a defensible refinement — measured pruned-vs-archival retention, regional cost heterogeneity, or corrected discounting — moves the banded

result ($\sim 22\text{--}29\%$ coverage, most blocks below $1\times$) to a qualitatively different conclusion, the paper’s central claim is falsified in its current form.

3. **Fees consistently exceed modeled long-term costs.** If sustained fee regimes (or re-measured 2017–2024 fee-peak years) show fees *covering* modeled costs in a stationary way, the partial-internalization reading collapses into “the market internalizes storage when it matters” — a different result, and the banded headline (§5) would be wrong.

Falsifiers of the framework (the RIR family and the “no single resource market” thesis):

4. **Resource-cost attribution is shown economically inappropriate.** If the planned attribute-pricing regression (companion `future-directions-v3.md` §2 Q2) shows the single fee price carries *no* persistence signal, then per-resource internalization ratios are category errors, not measurements — the RIR family premise fails even though SCCR-as-computed may stand.
5. **The pruned-vs-archival census shows the storage burden is avoidable at scale.** The $T = 10$ replicated-storage denominator assumes archival retention across the network. A measured pruning distribution showing most nodes avoid most of the burden (companion note `archival-vs-pruned-note.md`) would reframe SCCR as an upper bound on an avoidable cost — a genuine falsification of the externality reading, though not of the measurement.
6. **Measured response functions close the dynamic loop at or above $1\times$.** If node-entry/exit and fee-demand response functions (companion `future-directions-v3.md` §2 Q1) are measured and exhibit a stable fixed point at or above $1\times$, the persistent-partial-internalization equilibrium hypothesis is falsified.

We do not believe any of these falsifiers currently obtain; we list them so the reader can check, and so the framework is never mistaken for an unfalsifiable claim.

7.2 Assumption taxonomy — deliberate choices (Type C) vs. empirical risks (Type M)

(Added 2026-08-03, fourth-reviewer deliverable.) §7.1 draws a categorical line between two external outcomes — a **failed reproduction** (the number is wrong) and a **reproduced number with disagreed framing** (the number stands). That line is only enforceable if the paper’s own assumptions are pre-sorted into the same two buckets **before** the challenge arrives — not argued case-by-case under time pressure when an external review lands. Every material assumption in this paper therefore belongs to exactly one of two categories, stated here in advance:

The crisp test. A challenge is classified by what it does to the assumption’s current value, not by who makes it:

- **If a critic simply asserts a different value or a different modeling choice** ($T = 20$, marginal attribution, “validation matters more”) — there is no evidence the current value is wrong. That is **Type C territory: recorded as feedback** (community-feedback triage, `research/community-review-plan.md` §4), engaged in the next revision, and **the measurement stands**.
- **If the critic shows the current value or measurement is wrong** — with evidence, a reproduction mismatch, or better data (a complete node census, a corrected fee capture, a measured cost decomposition) — that is **Type M territory: a falsification candidate** under §7.1 falsifiers 1–3, and **the headline is at risk** until the discrepancy is reconciled.

Type C — “Assumption I chose” (deliberate modeling choice). A selection among defensible alternatives, made explicit and documented where it appears (§7 items 1–7). Not falsifiable by asserting a different choice; a reviewer preferring another selection records **feedback**, not

falsification. Reclassification to Type M requires showing the choice is internally inconsistent with the paper’s own stated method, or rests on an empirical premise that is wrong.

Type M — “Assumption that might be mismeasured” (empirical risk). An empirical quantity, measurement, or bound embedded in the model. Falsifiable: if the measurement is wrong, the conclusion may shift. A critic who shows the current value/measurement is wrong opens a **falsification candidate** — handled under §7.1 falsifiers 1–3; the headline falls until reconciled.

The blurry case, named honestly. Row 3 (C bundling) is the one assumption that sits on the line. The *decision to bundle* is a choice (Type C); the *claim that the bundle’s components are as measured* is empirical (Type M). Because the bandwidth leg (600) is the largest component, the storage-specific reading of the headline depends on a decomposition that is currently *assumed*, not measured. We therefore classify C bundling **leaning Type M** — the safe default: an external challenge to the decomposition gets the full falsification response, not the feedback bucket.

Why the taxonomy is stated in advance. Every assumption-level critique this paper receives will be routed by this table before it is answered: Type C critiques are engaged as feedback in the next revision (the reproduced number stands); Type M critiques are treated as falsification candidates until the evidence is reconciled (§7.1 falsifiers 1–3). This is the mechanism that keeps “reproduced the number, disagrees with framing” from silently becoming “the paper might just be wrong” — and keeps a genuine mismeasurement from being dismissed as a mere framing disagreement. The plan-of-record mirror lives in `roadmap.md` §12.

8 Economics and Related Work

8.1 Is this an externality?

Following standard environmental-economics usage (Pigou, 1920; Coase, 1960), a **negative externality** arises when a transaction’s cost is borne by parties who did not consent. Applied to Bitcoin: an inscription’s fee is paid by its creator; the long-term storage cost is borne by node operators who neither created the transaction nor were compensated for it. Two qualifications are stated honestly:

1. **Voluntary participation.** Node operators choose to run nodes (and may prune). The Pigouvian case is therefore weaker than for a physical externality — the correct framing is an *unpriced but avoidable* cost, not an imposed one. **Voluntary participation weakens the welfare interpretation, but not the measurement** — the ratio still quantifies what the fee market covers of a real, recurring cost borne by the storage-bearing class, whatever the operator’s freedom to exit.
2. **Measurement, not pricing.** This paper *measures* a ratio; it contains no price mechanism and proposes no policy.

8.2 Related work and novelty

- **Aronoff, Praizner, Sabouri (2026), arXiv:2604.17183** — structural VCG fee model; treats the mempool as a market for scarce block space; does not model storage externalities.
- **Liu, Fang, Cheung, Cai, Huang (2021), arXiv:2103.05866** — “*An Incentive Mechanism for Sustainable Blockchain Storage.*” Argues storage costs have “in general not been properly compensated by the users’ transaction fees.” **This is the closest prior work** and we acknowledge it directly: our contribution is not the observation (already argued in 2021), but a *measured, reproducible, Bitcoin-live-data quantification* (the SCCR) with a reconciliation of cost models and a knife-edge sensitivity bound. **We do not claim the observation**

is novel; we claim the measurement is.

- **Ethereum state-rent / gas-as-state-pricing** and **Sompolinsky–Zohar** are adjacent threads we build toward but do not model.

8.3 The efficient-markets objection (named rebuttal)

Objection: if block space is priced at the margin by a market that clears, then marginal cost is internalized by definition — the fee IS the price, so there is no externality to measure.

Rebuttal: this conflates two different horizons. The fee market prices *inclusion in the next block* — a static, congestion-clearing price with a ~ 10 -min horizon. The storage cost the SCCR measures is a *recurring* cost over an indefinite horizon (the data persists in every full node’s history). A market can clear for the short-horizon good (block space now) while not pricing the long-horizon good (permanent replicated storage). The paper’s measurement question is precisely whether the one-time congestion price also covers the recurring storage cost; §5 shows the empirical answer (banded ~ 22 – 29% coverage at $N=32K$). The efficient-markets framing is therefore not contradicted — it is *scoped*: it describes the short-horizon market, and the paper measures the long-horizon gap.

Designer-intent corroboration (primary sources). The scoped reading is not a reinterpretation — it matches the system’s own design intent. The whitepaper defines the fee solely as an incentive to create blocks: “*If the output value of a transaction is less than its input value, the difference is a transaction fee that is added to the incentive value of the block*” (§6), and storage is explicitly designed *around*, not priced: the 80-byte block-header estimate (“*80 bytes * 6 * 24 * 365 = 4.2MB per year*”) is dismissed with “*storage should not be a problem*” (§7) — a claim about block headers kept in memory for SPV, not full-chain replication, which is precisely the storage this paper measures. When Satoshi later moved to protect disk space, he used quantity control, not price: the block-size threshold should be kept “*lower as a circuit breaker*”, explicitly to “*limit the amount of wasted disk space in that event*” (BitcoinTalk post 441, Sep 2010), and the cap was designed to be raised by a block-height trigger, “*if (blocknumber > 115000) maxblocksize = largerlimit*” (post 485, Oct 2010). The efficient-markets objection therefore fails on the designer’s own terms: the single price was specified as an inclusion incentive; storage was handled — where it was handled at all — by quantity controls, never by price. (Full verification, exact sources, the apocryphal-quote flag, and a falsifiable-claims table derived from these primary texts: companion note `research/satoshi-primary-source-note.md`; the “node equilibrium self-corrects” claim maps to roadmap Q1.)

8.4 Terminology

This paper’s primary name for the metric is **Storage Cost Coverage Ratio (SCCR)** — a fee-to-modeled-cost comparison, *not* a coverage ratio in the insurance/solvency sense. The broader economics family generalizes it: **Cost Internalization Ratio** is the family name for “the share of a long-lived resource cost internalized by the one-time fee,” of which SCCR (storage) is the first measured member (Metric #1 of the RIR family; companion `future-directions-v3.md` §2 Q3). We use SCCR / Storage Cost Coverage Ratio as the established name throughout; Cost Internalization Ratio appears here once to locate the family.

9 Conclusion

Whether the fee market internalizes long-lived resource costs is an **open empirical question**, and this paper contributes the first reproducible measurement using a **primary-source lower-**

bound census ($\geq 32,000$ known addresses via Bitcoin Core `getnodeaddresses`): a storage-cost-internalization ratio of **$\sim 0.22\text{--}0.29$ at $N=32K$** (167–156 blocks, dated captures), with **$\sim 98.7\text{--}100\%$ of sampled blocks below the $1\times$ threshold** at that node count. Because the replication factor N is the dominant input and is only bounded ($\geq 32K$; independent estimates 10K–100K), the honest headline is a **range**: fees currently cover roughly **7% to 71%** of the modeled 10-year replicated storage cost depending on the true node count, with the *direction* (most blocks below $1\times$ under any $N \geq \sim 32K$ census) robust to every correction made in this paper’s review. **Status**: external reproduction is pending (D5, the only submission blocker); every figure is stated to the precision the current evidence licenses, and the N-band range carries the dominant uncertainty.

10 Future Directions (v3.0) — the agenda, in a companion

The paper’s measurement question is answered in §5; its falsifiers are §7.1. The program’s forward agenda — the v3.0 economic-dynamics questions, their first answers, and the cross-chain generalization — **deliberately lives outside this core paper** in the companion `research/future-directions-v3.md` (plan-of-record: `roadmap.md` §8/§9/§11). This keeps the paper a *measurement*, not a research-program pitch. The eight v3.0 questions (headline model output at the live baseline, $SCCR = 0.2228$ at $N=32K$, $T=10$, $C=\$925/\text{yr}$):

#	Question	Headline model output (live baseline)
1	Storage horizon: $SCCR$ at $T = 5, 10, 20, 30, 50$ yr	0.446 / 0.223 / 0.111 / 0.074 / 0.045 (inverse-linear)
2	Storage $10\times$ cheaper: $C = \$92.5/\text{yr}$	$SCCR = 2.228$ — the gap flips sign
3	$BTC = \$500,000$	$SCCR = 1.768$; average crosses $1\times$ at $\sim \$283K$
4	Fees sustained at 100 sat/vB for 5 yr	$SCCR \approx 11.2$ (22.4 at $T=5$); crosses $1\times$ at ~ 9 sat/vB
5	Lightning moves 90% of payments off-chain	Two-sided; net effect open
6	Nodes double / triple: $N = 32K \rightarrow 64K \rightarrow 128K$	$0.223 \rightarrow 0.111 \rightarrow 0.056$ (inverse-linear)
7	Can $SCCR$ reach 1 without protocol changes?	Historically yes: above $1\times$ in 2017–2024 fee-peak years
8	What is the equilibrium?	Open. The model measures a ratio; it does not close t

Open-question honesty. Nothing in this section claims the fee market *will* internalize these costs, nor that it *should*. The claim is narrower: the question is now well-posed, the drivers are identified, and each driver’s lever on the ratio is measured.

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#	Assumption	Type	Why it's that type	What would reclassify or falsify it
1	Storage horizon $T = 10$ yr	C	A selection within a defensible range (pruning shortens retention, permanent storage extends it, §7 item 3); sensitivity disclosed (§5.3: $T = 5/10/15 \rightarrow 0.446/0.223/0.149$), direction robust	A measured network-wide retention distribution showing effective retention is an order of magnitude shorter — reframes the externality reading (falsifier 5); does not make $T = 10$ an error
2	Replication factor $N = 32K$	M	An empirical bound from the primary-source census (addrman cap at 32,000 known addresses), explicitly <i>not</i> a complete enumeration (§5.4); true set estimated 10K–100K; SCCR $\propto 1/N$	A defensible complete census showing the true reachable set differs materially from 32K in a direction that moves the ~ 0.07 – 0.71 band; the “ ~ 99 – 100% below $1\times$ ” claim already breaks at $N \approx 49K$ (§5.4)
3	C bundling ($C = C_{\text{storage}} + C_{\text{bandwidth}} + C_{\text{misc}}$)	M (leaning — the blurry case)	Bundling is presented as a simplification (Type C-like), but it embeds an implicit empirical claim that the components are as measured — and the component sum is bandwidth-dominated ($600 + 166.67 + 157.68$, §4.1), so the <i>storage</i> reading of the headline is fragile if the decomposition is wrong	A measured node-cost decomposition (operator surveys, hosting bills) showing the storage share is materially different, or a double-counted component — reframes the headline from “storage-and-hosting coverage” (§4.1) and re-bands the ratio
4	Average block size $B_{\text{block}} = 1.5\text{MB}$	M (narrow surface)	An empirical quantity from the fee-history capture — but §5.3 shows SCCR is invariant to B (B cancels), so a wrong value changes the per-byte presentation (c_b), not the headline ratio	A material mis-statement of the capture (wrong byte basis in the fee history) propagating through $c_b(t)$; a data-quality flag, not headline-moving under the current spec
5	“Fees paid” definition ($\text{fee_USD} = \text{avgFees} \times \text{USD/BTC}$; subsidy excluded, §4.1)	M	The numerator is a measurement from the live capture (mempool.space 24h block-fee history); the definition is explicit but the value is empirical	A reproduction mismatch on the fee side, a corrected capture, or evidence the conversion/attribution is wrong — direct falsifier 1/3 territory: the number falls until reconciled
6	Storage-first sequencing	C	A program ordering, explicitly “the first measurable one,” not “the most important” (§7 item 7)	Only evidence that storage is <i>not</i> reproducibly measurable (falsifier 1) touches it; asserting validation/UTXO is more important is feedback, not falsification
7	Average-vs-marginal attribution	C	A documented choice (the $164\times$ denominator gap is disclosed; both branches reported, §4.2); average answers the accounting question, marginal the optimization question	Showing the marginal object is the only economically correct one for the externality claim — the paper reports both, so this is engagement, not error
8	Homogeneous node cost structure	C (with an empirical seam)	The single-scalar structure is a deliberate simplification, documented (§7 item 2); heterogeneity is a refinement direction, not an error	The <i>value inside</i> the structure ($C = \$925/\text{yr}$) is empirical — a defensible regional refinement that moves the band is already named falsifier 2; the structural